

Course Outline

Programming Fundamentals

Text Books:

- A. E. Balagurusamy - Computing Fundamentals and C Programming-Mc Graw Hill India (2018)

Reference Books:

- B. Programming with C schaum outline series

Course Description:

The following topics are part of the course; however, the ordering may be different and some topics may not be covered due to time.

Introduction to computers & programming (memory, programs, input, output, binary, ASCII Chart).

Program using printf function (escape sequence \n, \t, \r, \b)

Variable, memory, input, data types

Print specifier %d, %c, %f, %s

Program using scanf function for input

Arithmetic Operators (+, -, *, /, %), operator precedence, integer division & complete division

Examples of complete, integer division and remainder

Relational Operators ==, !=, >, >=, <, <=

Decision Making using if statement

If-else chaining, Nested if

Logical operators for composite conditions &&, ||

Loop constructs while & for, infinite loop

Conditional & counting loop

Nested loop

User define functions

Array (Single dimension)

Loop and arrays

Two-dimensional arrays

Passing arrays to functions

Pointers

Dynamic memory allocation

Bit-wise operators &, |, ~, <<, >>

Structures